BSIM3021 Web Development, Users and Management

Lecture 4: Usability & User-centered Evaluation

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The University of Hong Kong

Acknowledgement

- I would like to thank Prof. Xiao Hu, Dr. Alvin Kwan, and Prof. Shahla Ali for kindly sharing their previous course materials for my adaptation.
- I would also like to thank other dedicated educators and the like for making their resources (e.g., MIT Open Courseware) available online for public access and adaptation.

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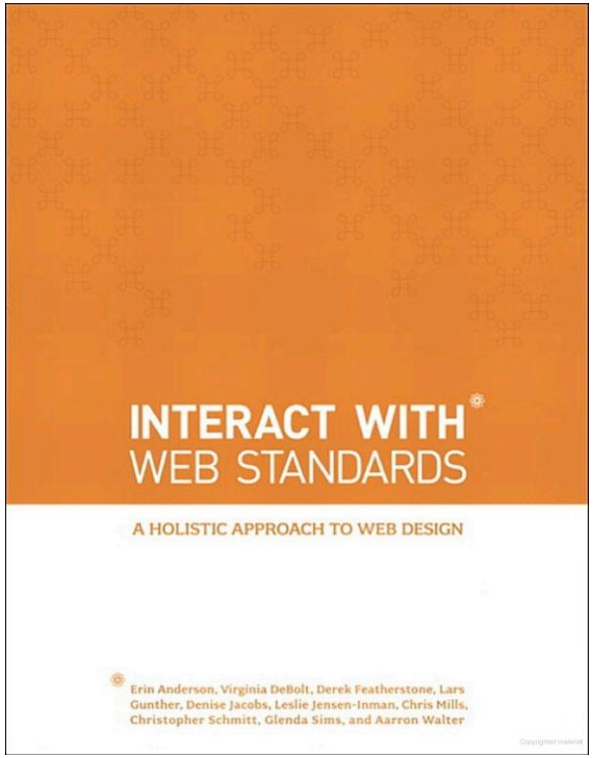
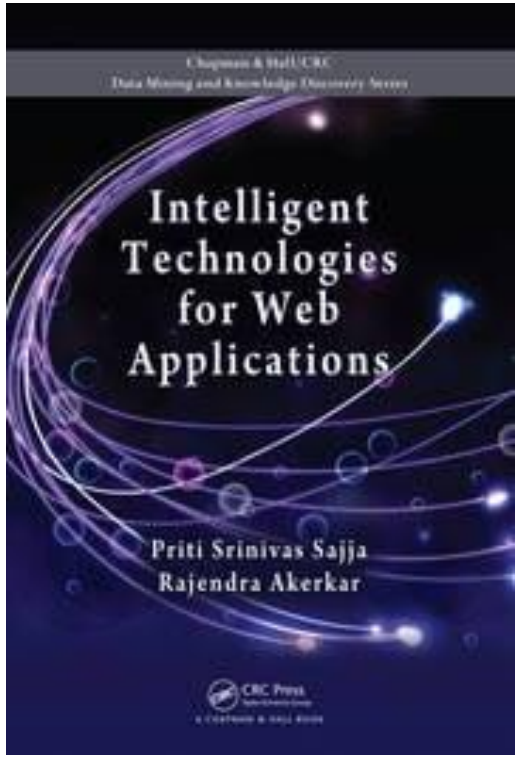
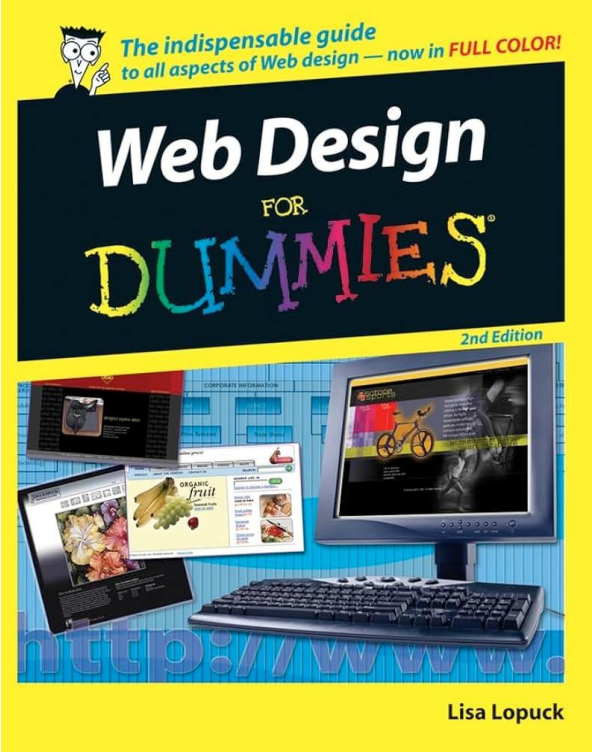
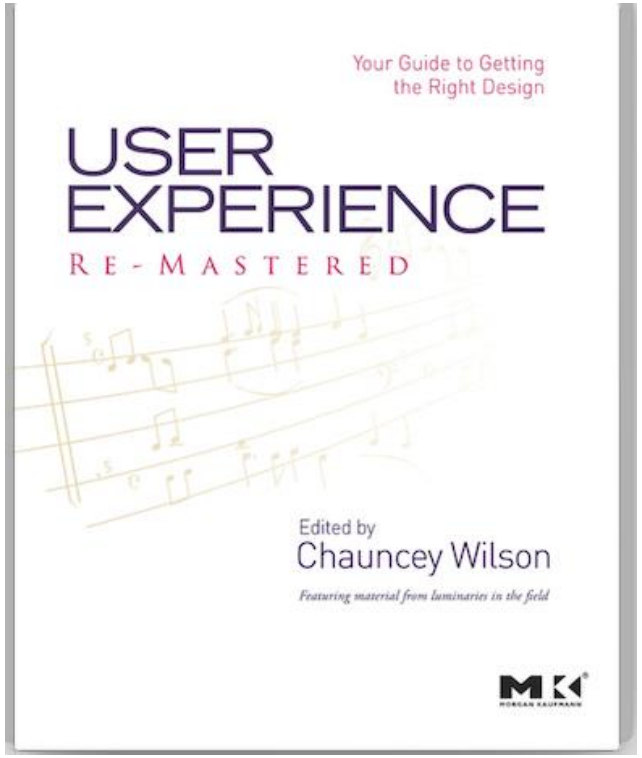
Disclaimer

In addition to adapting contents from colleagues, this lecture is largely composed of and/or inspired by contents and materials from









… and the **collective wisdom** from the Internet. Feel free to raise alternative ideas and opinions at any point.

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Outline

- **Recap**
- Usability
- User-Centered Evaluation
- Preparation for assignments

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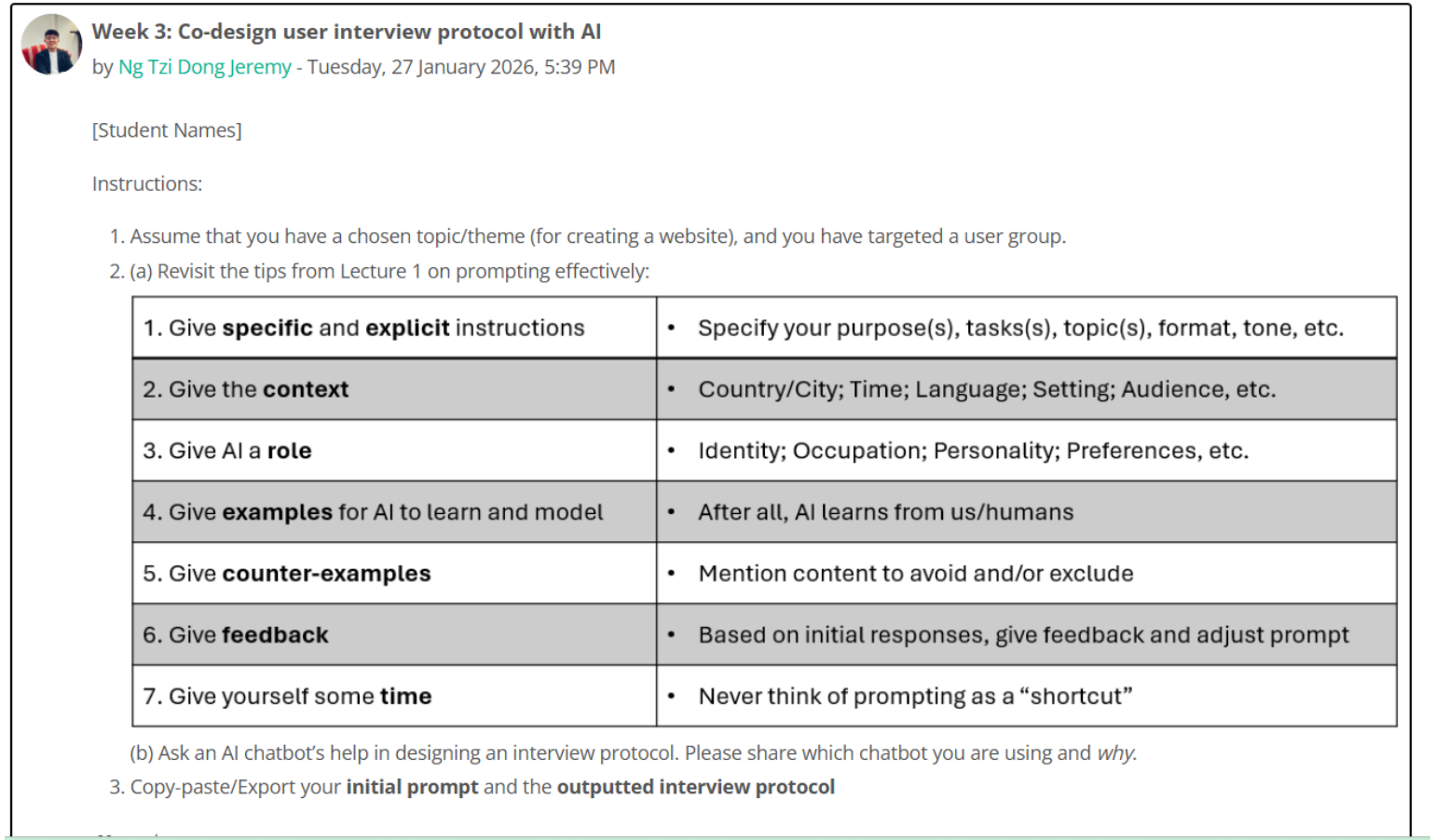
Feedback on Week 2 task: AI-assisted interview protocol design



Week 3: Co-design user interview protocol with AI

by Ng Tzi Dong Jeremy - Tuesday, 27 January 2026, 5:39 PM

[Student Names]

Instructions:

1. Assume that you have a chosen topic/theme (for creating a website), and you have targeted a user group.
2. (a) Revisit the tips from Lecture 1 on prompting effectively:

1. Give **specific** and **explicit** instructions	• Specify your purpose(s), tasks(s), topic(s), format, tone, etc.
2. Give the **context**	• Country/City; Time; Language; Setting; Audience, etc.
3. Give AI a **role**	• Identity; Occupation; Personality; Preferences, etc.
4. Give **examples** for AI to learn and model	• After all, AI learns from us/humans
5. Give **counter-examples**	• Mention content to avoid and/or exclude
6. Give **feedback**	• Based on initial responses, give feedback and adjust prompt
7. Give yourself some **time**	• Never think of prompting as a "shortcut"

(b) Ask an AI chatbot's help in designing an interview protocol. Please share which chatbot you are using and *why*.

3. Copy-paste/Export your **initial prompt** and the **outputted interview protocol**

https://moodle.hku.hk/mod/forum/discuss.php?d=1007906

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- Recap
- **Usability**
  - **What is Usability**
- User-Centered Evaluation
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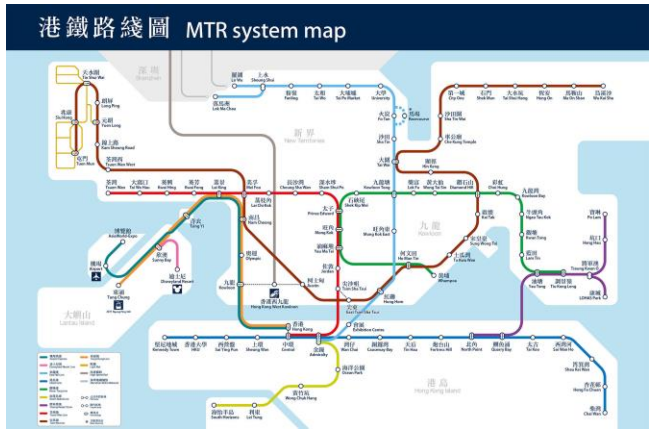
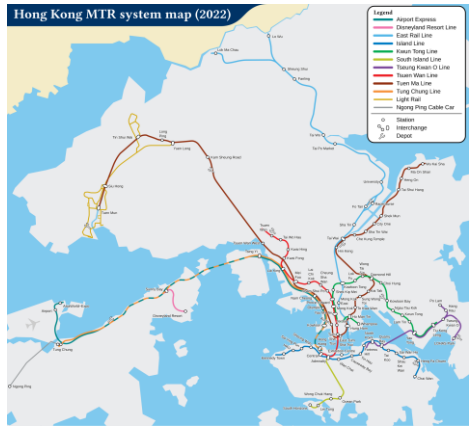
What is Usability?

- Which MTR map has the “best usability”? Why?



(A)

(B)



(C)

(D)

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Usability

- **Usability**: a quality attribute that assesses how easy user interfaces (UI) are to use
  - Not a single, one-dimensional property of a UI
- Five components:

https://www.nngroup.com/articles/usability-101-introduction-to-usability/

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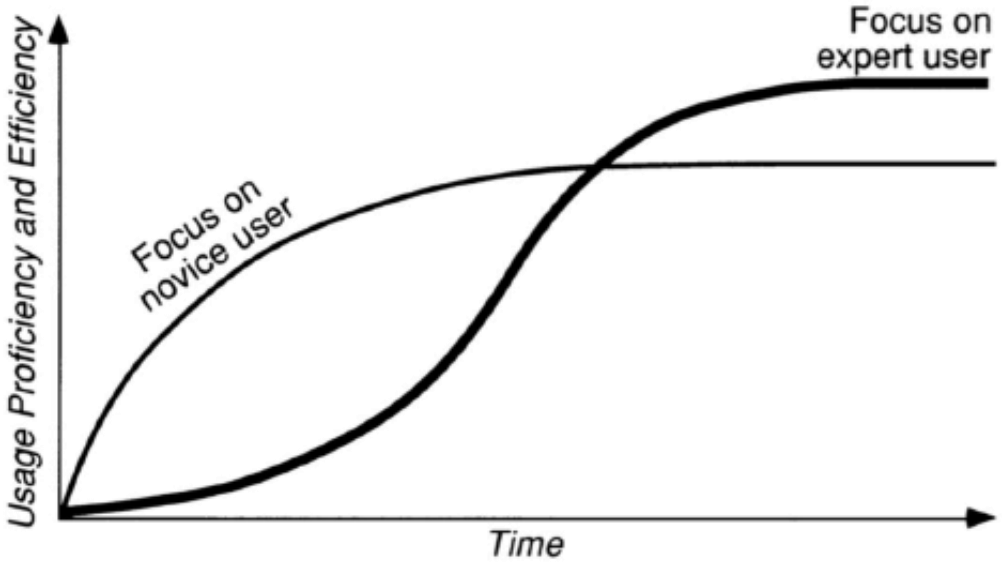
Usability #1: *Learnability*

How easy is it to accomplish basic tasks the first time they encounter the design?

- Practically, all user interfaces have **learning curves** that start out with users not being able to do anything (i.e., zero efficiency)

- How to measure *learnability*: given a user having no experience with a website (i.e., **novices**), we measure the time it takes for them to reach a specified level of proficiency
- No need for the user to learn a complete interface fully *before* using it



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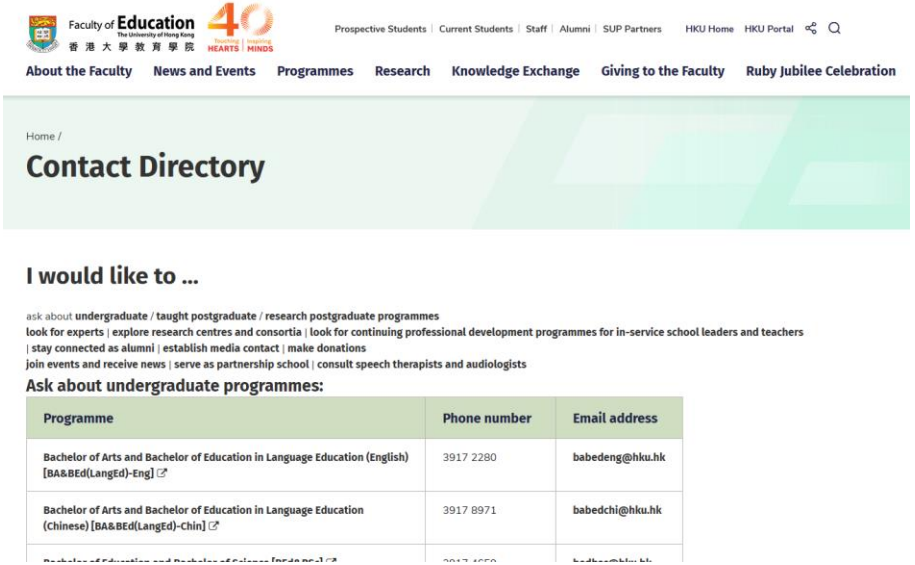
https://www.nngroup.com/articles/flexibility-efficiency-heuristic/

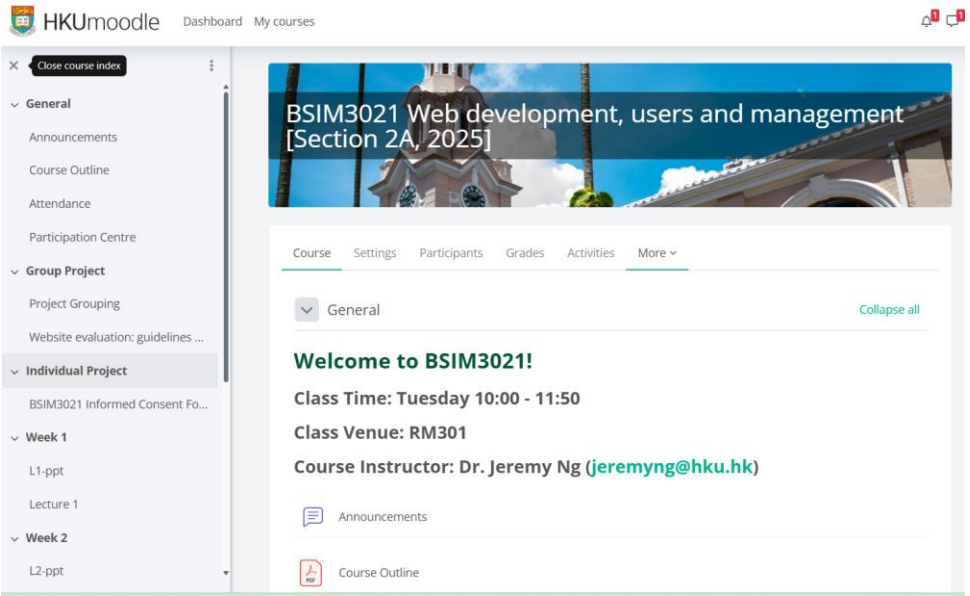
Usability #2: *Efficiency*

Once users have learned the design, how quickly can they perform tasks?

- One of the usability heuristics: To *prioritize* flexibility and efficiency
- Example: Use **shortcuts** – unseen by the novice user, and speed up interaction for the expert user



https://web.edu.hku.hk/contact-directory



Shortcuts as part of the *navigation menu*

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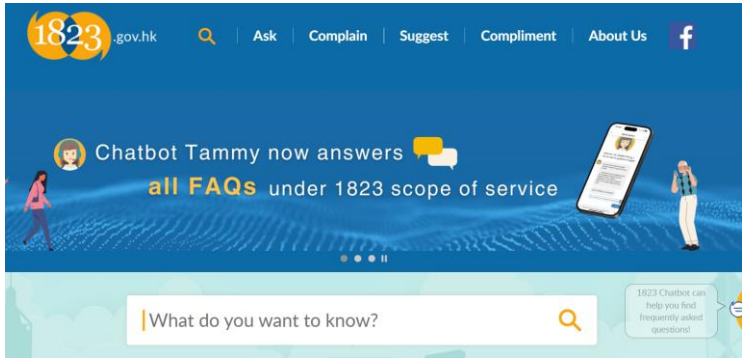
Usability #3: *Memorability*

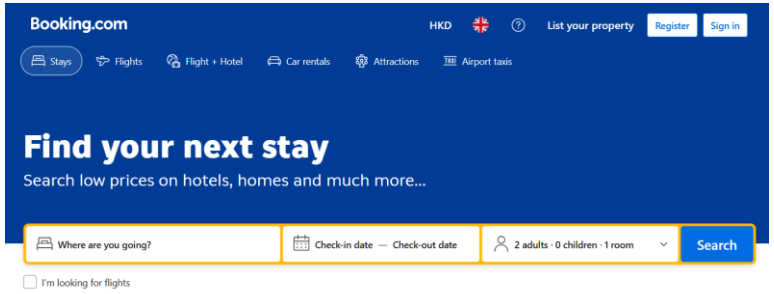
When users return to the design after a period of not using it, how easily can they re-establish proficiency?

- UI learnability is relatively less tested thoroughly, when compared with other usability attributes
  - Can be tested by assessing *how much* a user remembers the **commands** on the website

- Examples: Websites we infrequently visit, but their *memorability* is important:



https://www.1823.gov.hk/en



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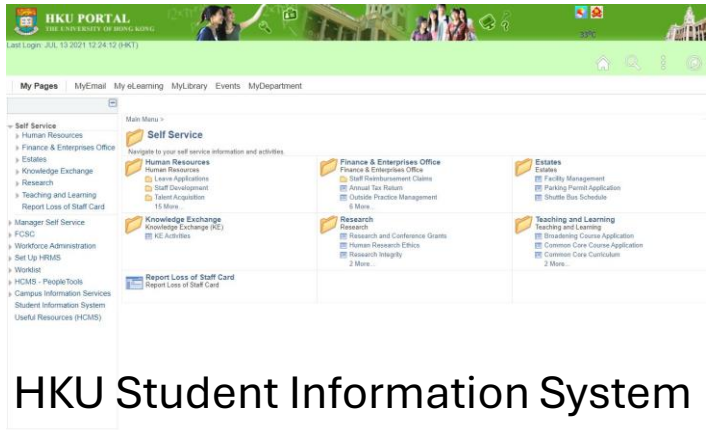

HKU Student Information System

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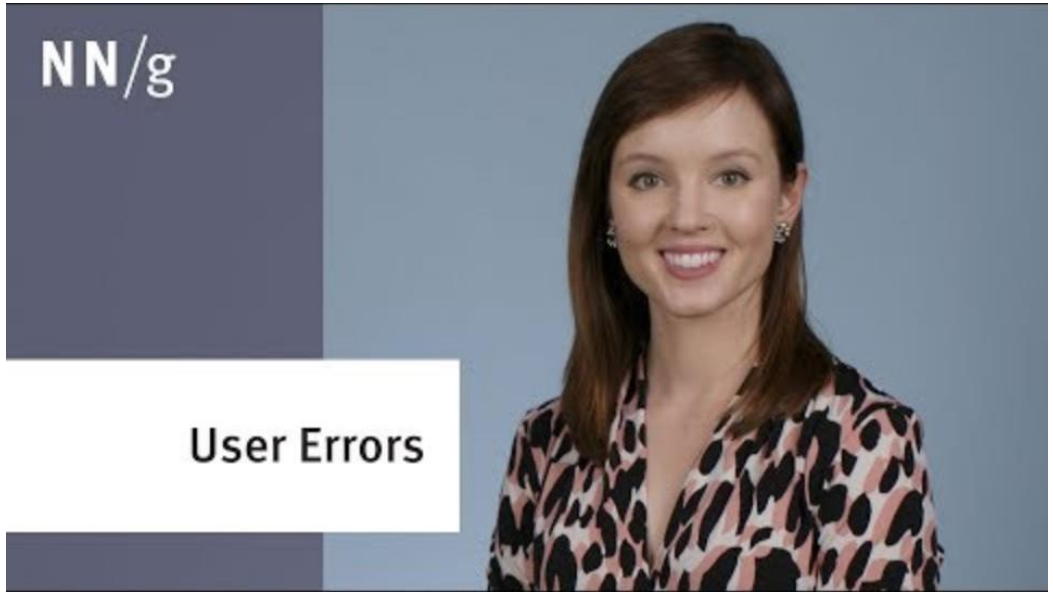
Usability #4: *Errors*

How many errors do users make, how severe are these errors, and how easily can they recover from the errors?

- **Errors** = any action that does not accomplish the desired goal
  - Minor errors: do not slow down users' browsing significantly and can get corrected
  - Some errors are *catastrophic*, e.g., not discovered by the user, destroying user's work, etc.



https://www.youtube.com/watch?v=cCun-ReLTFI

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Usability #5: *Satisfaction*

How pleasant is it to use the design?

- **Subjective satisfaction**: important in all user-facing systems, e.g., websites, apps, games, immersive environments etc.
  - Complemented/supplemented by *objective* measures, e.g., pupil dilation, heart rate, blood pressure, EEG, etc.

- Measuring satisfaction is easy AND difficult
  - Methods: **Survey**, **Interviews**, **Focus Group**, etc.
  - *When* to conduct? *What* to ask users? What *not* to ask? For *how long*?
  - Often measured together with other usability attributes

System Usability Scale (SUS) UXtweak

1. I think that I would like to use this system frequently.	1	2	3	4	5
2. I found the system unnecessarily complex.	1	2	3	4	5
3. I thought the system was easy to use.	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this system.	1	2	3	4	5
5. I found the various functions in this system were well-integrated.	1	2	3	4	5
6. I thought there was too much inconsistency in this system.	1	2	3	4	5
7. I would imagine that most people would learn to use this system very quickly.	1	2	3	4	5
8. I found the system very cumbersome to use.	1	2	3	4	5
9. I felt very confident using the system.	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this system.	1	2	3	4	5

https://blog.uxtweak.com/system-usability-scale/

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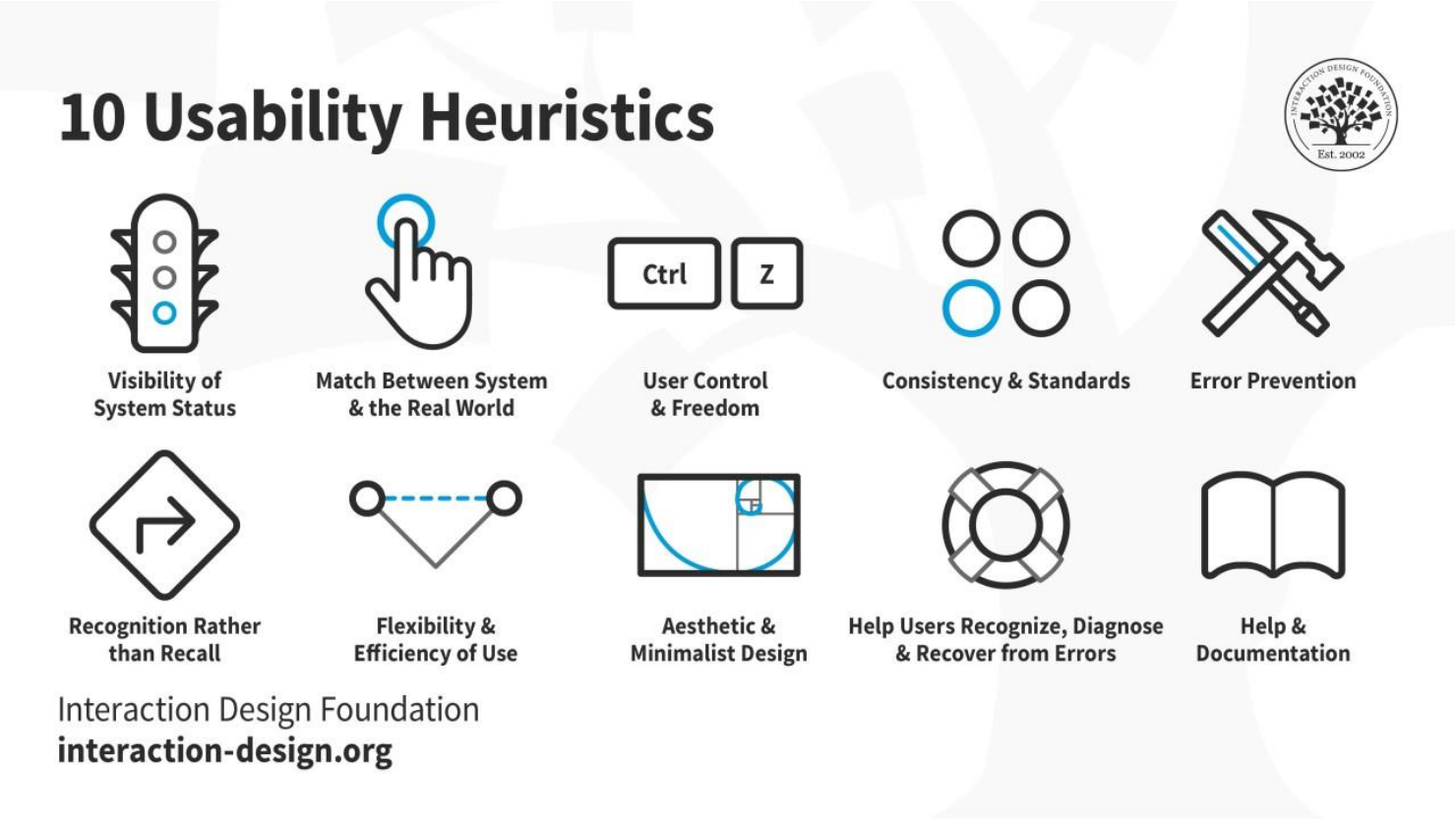
10 Usability Heuristics for UI Design

- There are 10 general principles for designing user interface of systems, including websites. They are *rules-of-thumb* but not specific *guidelines*.



https://www.nngroup.com/articles/ten-usability-heuristics

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In-class activity: Usability Evaluation (part 1)

Week 4: Usability Evaluation (part 1)

by Ng Tzi Dong Jeremy - Tuesday, 3 February 2026, 6:36 PM

[Please input names of all group members.]

1. Work in a group of 3 – 4 people.
   (You need to form a project group very soon.)
2. Explore the entire website: https://umag.hku.hk/
3. Evaluate its **usability** in terms of three of the 10 Usability Heuristics:
   https://www.nngroup.com/articles/ten-usability-heuristics

   Hint: "Evaluate" = *critically* comment on the website, both positively and negatively

54 words

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https://moodle.hku.hk/mod/forum/discuss.php?d=1011312

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  - What is Usability
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Why is Usability important for a website's survival

Improve user engagement

SEO benefits

Reduce costs in the long run

Improve web accessibility

If the website is difficult for the users to use, they **leave.**

If the homepage fails to clearly state what the website offers to the users, they **leave.**

If the website's info is hard for the users to read, they **leave.**

If users get lost on the website, they **leave.**

https://www.nngroup.com/articles/usability-101-introduction-to-usability/

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  - **How to improve Usability**
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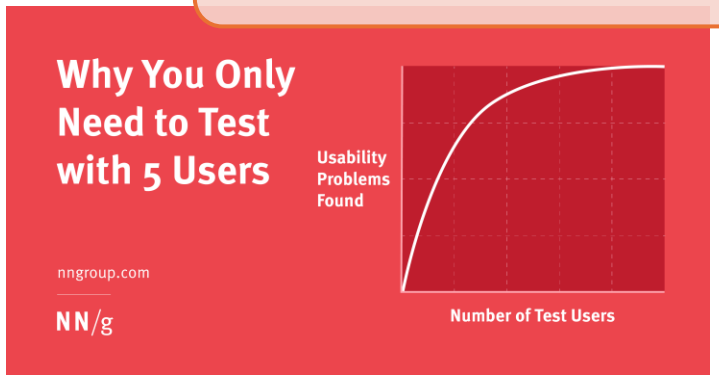
How to improve the Usability of your website: *Think-aloud protocol*

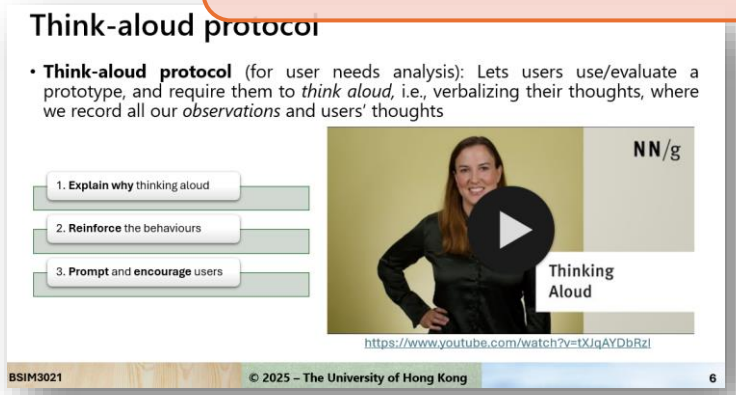
Identify your *representative* users



Ask users to perform *representative* tasks

- **Realistic**
- **Actionable**
- **Don't give clues**

Observe users' behaviours: "***shut up and let users talk***."



https://www.nngroup.com/articles/recruiting-test-participants-for-usability-studies/

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How to improve the Usability of your website

In addition to **Nielsen's 10 Usability Heuristics,** there are many other web design principles that you can follow, for improving the usability of your website:

Don't make users think

- Be intuitive, simple, obvious, and self-explanatory
- Consistent design
- Follow conventions

Don't waste users' patience

- ***2-second law***
- ***Hick's Law***
- ***3-click rule***

Manage to focus users' attention

- **Scan paths**
- ***7±2 principle***
- ***Fitts' law***
- ***80/20 rule***

Write effectively

- ***Inverted pyramid***

https://www.smashingmagazine.com/2008/01/10-principles-of-effective-web-design/

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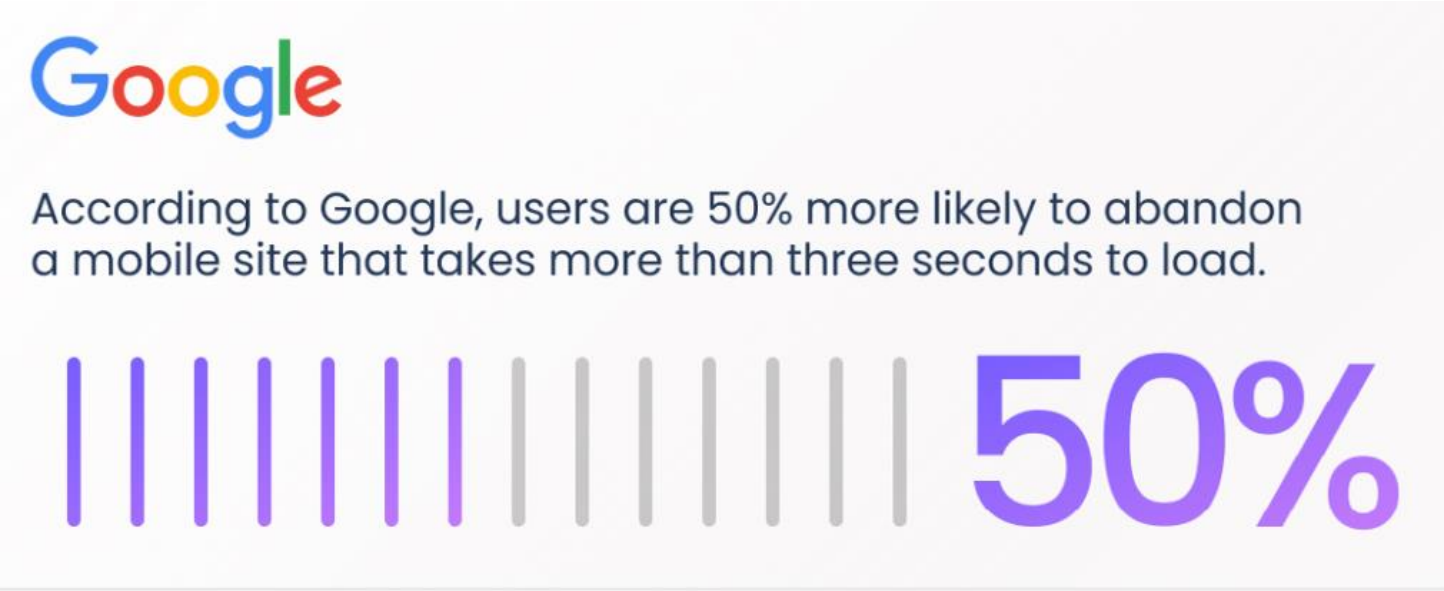
2-second law

- A website user should not need to wait for more than 2 seconds
- Findings of psychology research studies:
  - Students can only focus on a lecture for 10 – 15 mins
  - Digital device users have an **average task attention duration** of 47 seconds



https://venturebeat.com/business/the-impact-of-the-2-second-rule-why-website-speed-matters-more-than-you-think/
https://time.com/6302294/why-you-cant-focus-anymore-and-what-to-do-about-it/

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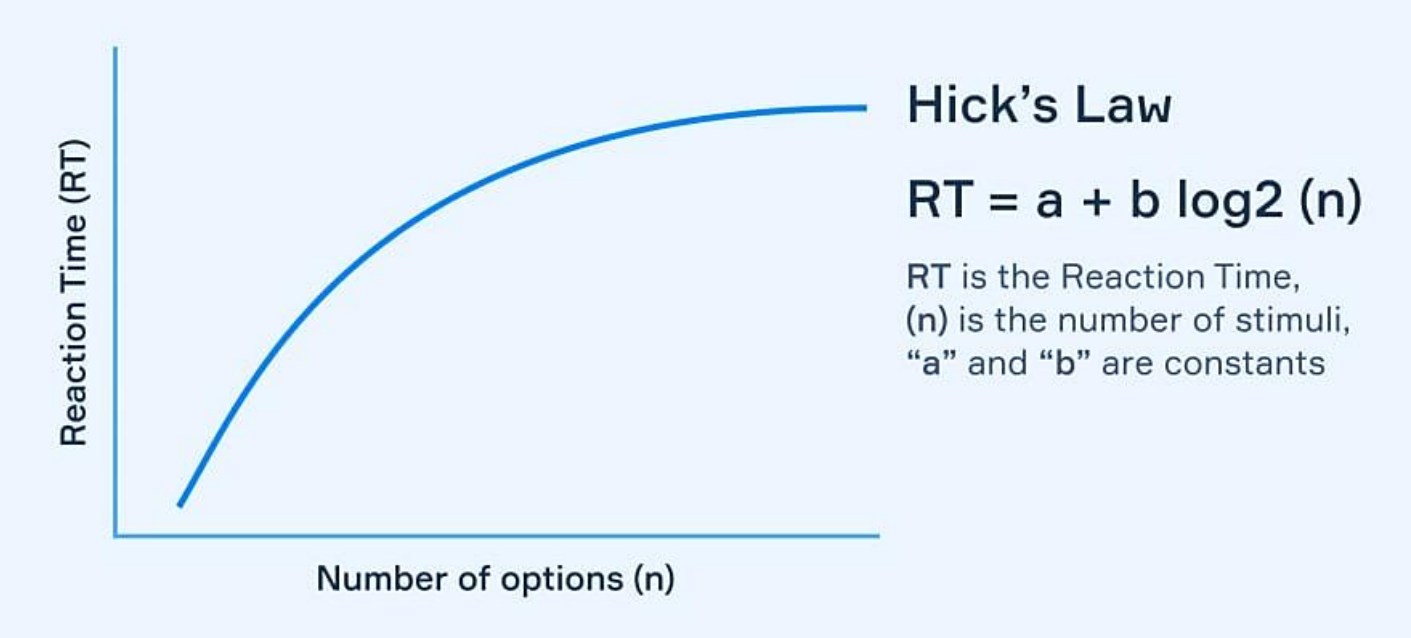
Hick-Hyman Law

- **Hick-Hyman Law**, a.k.a., **Hick's Law**: The more options, the more time it takes (i.e., the more difficult) to make a decision
- Implication for web designers: Provide only essential options on UI



https://dev.to/anasnmu/hicks-law-ijc

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The Three-click Rule: a debatable idea

- A rule-of-thumb that states that a user should always be able to find their desired information ***within* three clicks**
- No official/research evidence as support – but can be of reference
- Recommendations:
  - Avoid vague, overly broad terms
  - Avoid *multilevel hierarchical* dropdown menus

https://tillerdigital.com/glossary/3-click-rule/

https://www.nngroup.com/articles/3-click-rule/

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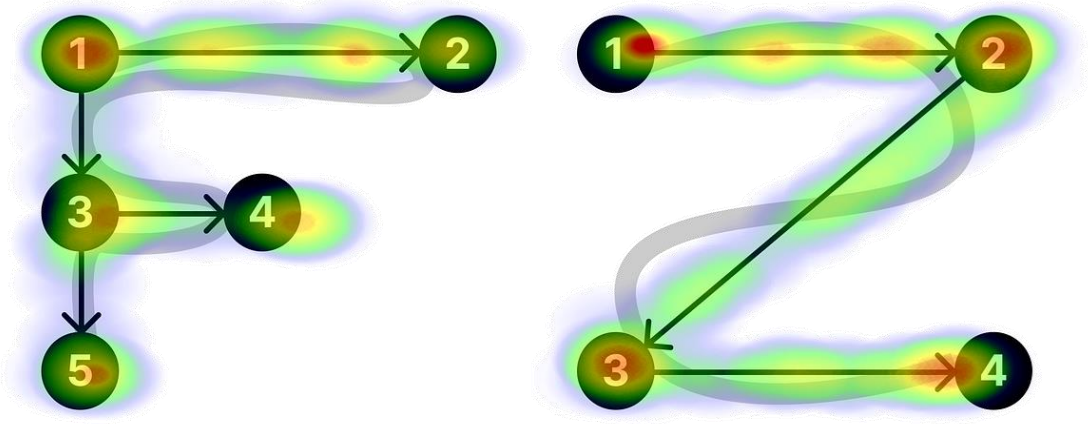
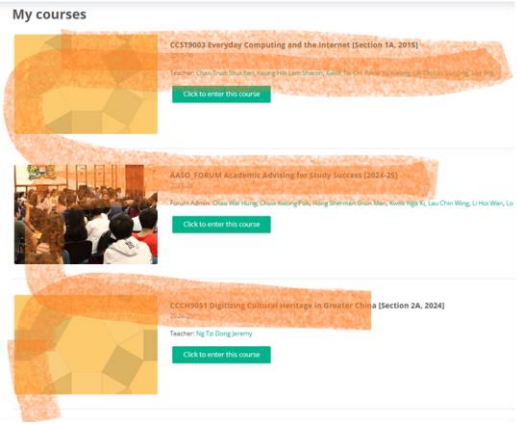
Common Scanpaths: *F Pattern* & *Z Pattern*





- Frequent patterns: **first lines** of text and **first few words** on the *left* of each line of the text receive more visual attention
- Implications for web designers: importance of **headings** and conciseness

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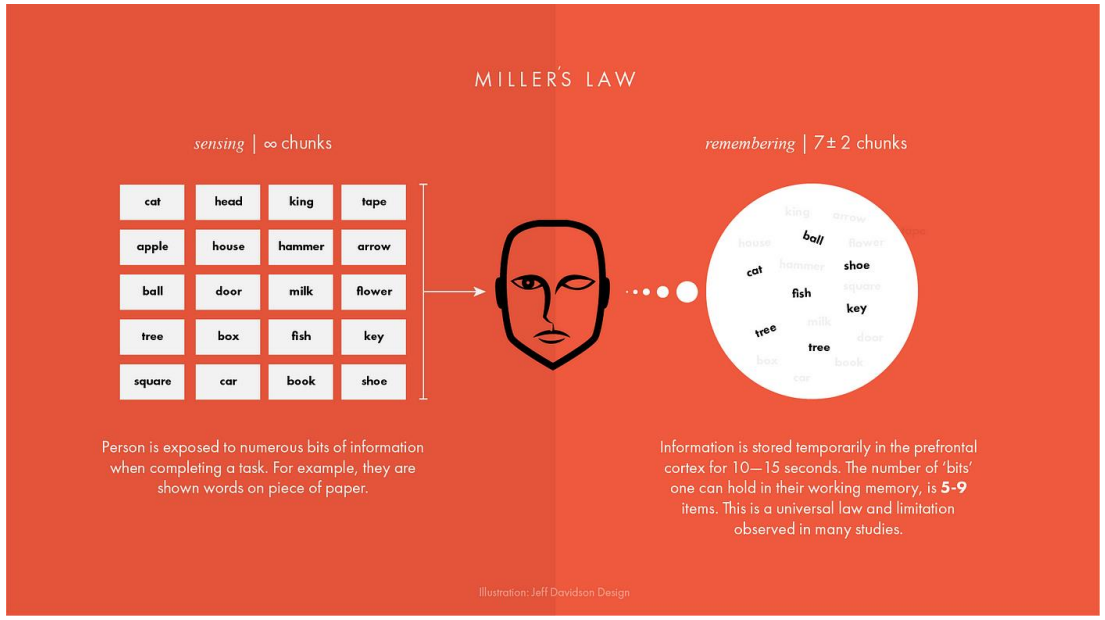
7±2 principle (“Magic number 7”)

- a.k.a. Miller’s Law, a *cognitive psychology* principle: The average person can only hold about 7(plus or minus 2) items in their working memory at a time

- Implications for website UI design: present information in **small chunks** that users can more easily remember, e.g.,
  - Show essential information first
  - Use clear and concise labels
  - Use visual cues and aids (e.g., icons)
  - Group related items on a menu



https://medium.com/design-bootcamp/the-7-2-rule-the-real-science-behind-millers-law-that-will-shock-you-your-deepest-memory-4a35be25bb3b

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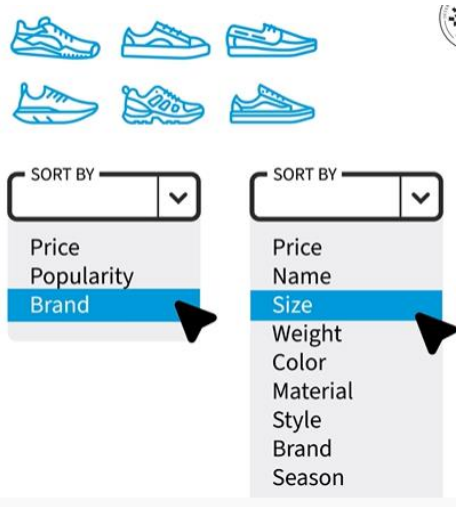
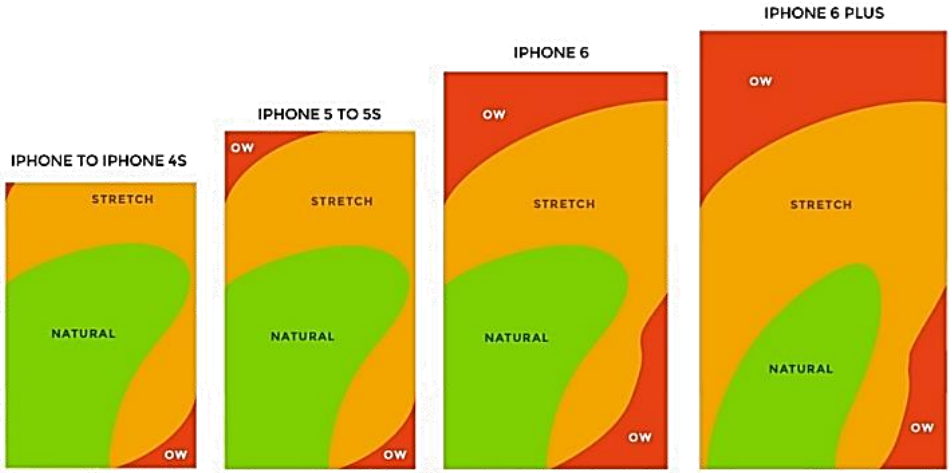
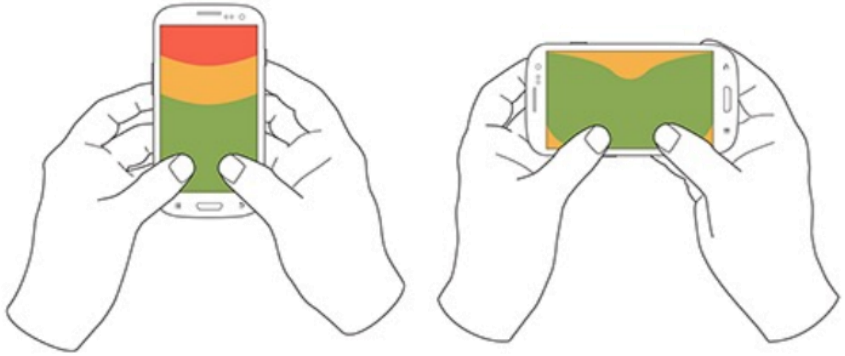
Fitts' law

- Fitts' law: the time needed for a pointer (e.g., mouse cursor) to reach a target depends on the distance to and size of the target
  - Big and near objects are easier to click
  - Small and far objects are more difficult to click
- Examples:







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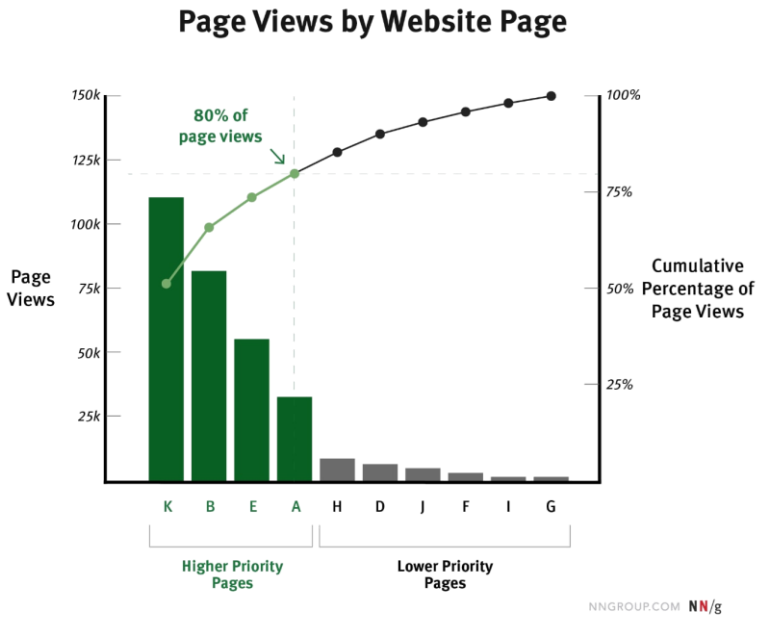
80/20 rule

- a.k.a. the *Pareto principle*: 80% of the results come from 20% of the cause
  - Only a hypothetical mathematical idea

- Examples:
  - 20% of websites capture 80% of web traffic.
  - 20% of customers generate 80% of a company's revenue.
  - 20% of software bugs contribute to 80% of computer crashes.
  - 20% of apps in our phones occupy 80% of our screen time.
- How to apply it in web design:
  - Identify the pages on your website with the highest page views
  - Prioritize the vital 20%
  - Optimize the less-important 80%



https://webdesignerdepot.com/the-8020-rule-applied-to-web-design/
https://www.nngroup.com/articles/pareto-principle/

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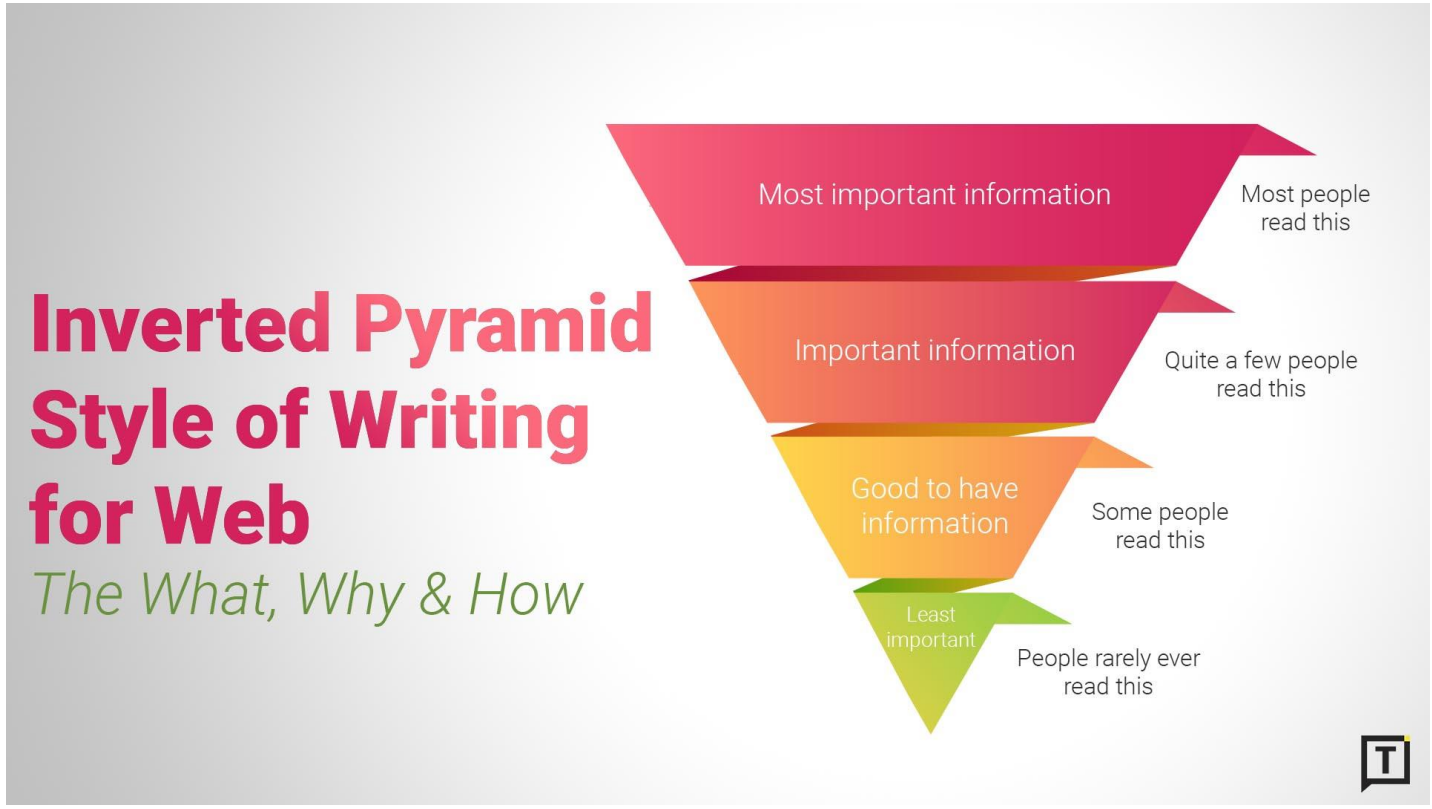
Inverted pyramid

- Putting the most important information at the beginning of a webpage



https://typewriter.media/2020/09/inverted-pyramid-style-writing/
https://www.surreycc.gov.uk/website/writing-guide/writing-web-pages-in-the-inverted-pyramid-style

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In-class activity: Usability Evaluation (part 2)

Week 4: Usability Evaluation (part 2)

by Ng Tzi Dong Jeremy - Tuesday, 3 February 2026, 6:45 PM

[Please input names of all group members.]

1. Work in a group of 3 – 4 people.
   (You need to form a project group very soon.)
2. Continue evaluating the entire website: https://umag.hku.hk/
3. Evaluate its **usability** in terms of:
   (a) *Hick's Law*
   (b) *Common scanpaths*
   (c) *7+/- 2 principle*

   Hint: "Evaluate" = *critically* comment on the website, both positively and negatively

58 words

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https://moodle.hku.hk/mod/forum/discuss.php?d=1011315

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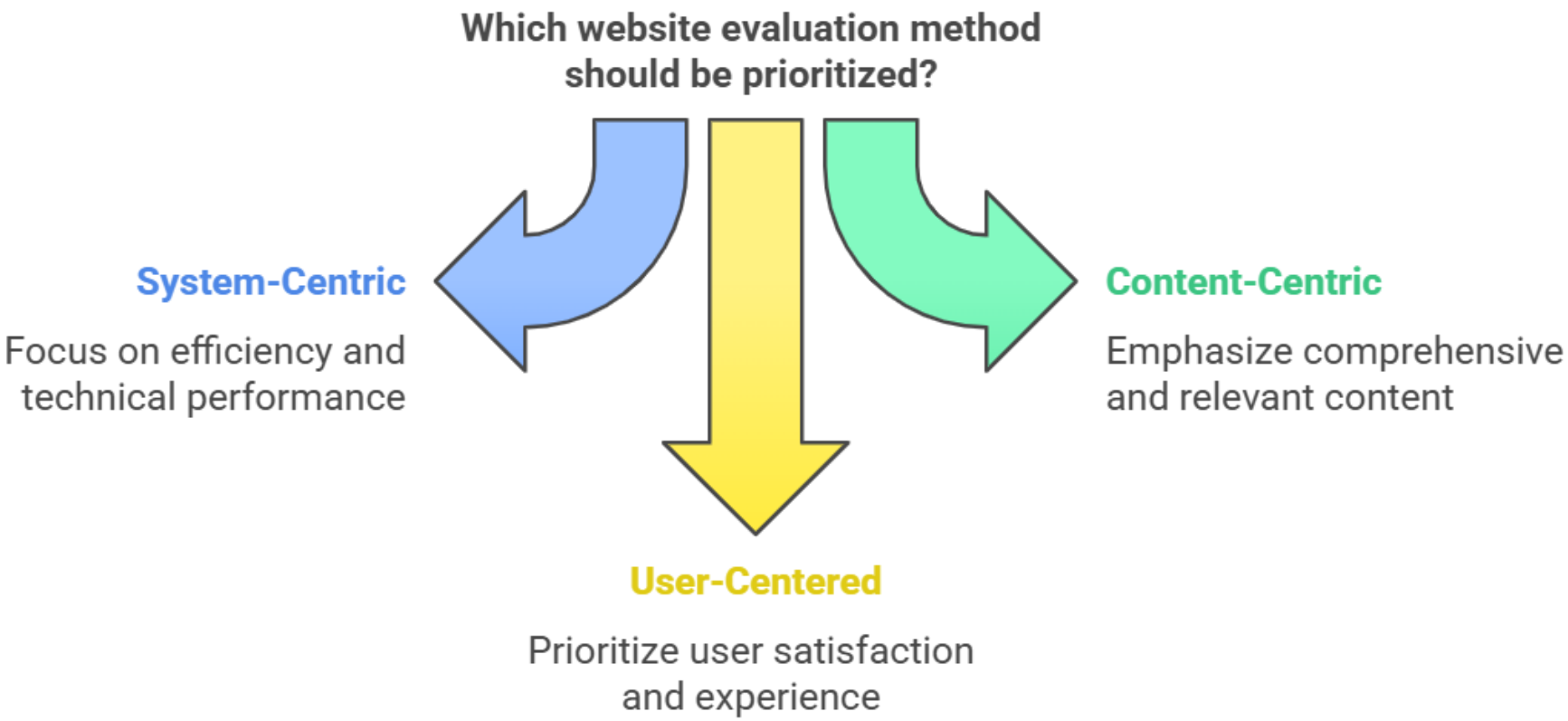
Three classes of website evaluation



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Methods of User-Centered Evaluation

- Largely the same as those for user needs analysis/assessment!

Survey (See Lecture 3)	Interview (See Lecture 3)	Focus group	Think-aloud (See Lecture 3)	Task analysis
Usability evaluation	Heuristic evaluation	Storyboarding	Prototyping	Log analysis
Content analysis	Experiments	Case study	Ethnographic analysis	Expert review

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5-second Usability Test

https://www.youtube.com/watch?v=X0FG0jCqLYQ

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Task Analysis

- **Task analysis**: systematic study of how users complete tasks to achieve their goals. It ensures our websites are designed to efficiently and appropriate support users' goals
  - The task should be an **observable activity** and with both a **starting point** and an **end point**
  - Also helps study users' *learning process* when interacting with a website

https://www.nngroup.com/articles/task-analysis/

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Log analysis

- ***Event/Activity logs***: provide a detailed *time-stamped* view of user interactions with a software system (e.g., a website)
  - Often *automatically* recorded by (and buried in) the system
- Typically, a website's event logs track what links/fields the users use, what errors occur, what content user inputs into specific fields, etc.
- Example: Moodle logs

Time	User full name	Affected user	Event context	Component	Event name	Description	Origin	IP address
27 January 2026, 10:31:19 PM	Ng Tzi Dong Jeremy	-	Course: BSIM3021 Web development, users and management [Section 2A, 2025]	System	Course viewed	The user with id '109125' viewed the course with id '126503'.	web	[redacted]
27 January 2026, 5:45:24 PM	Ng Tzi Dong Jeremy	-	Forum: Participation Centre	Forum	Discussion viewed	The user with id '109125' has viewed the discussion with id '1007906' in the forum with course module id '3997247'.	web	[redacted]
27 January 2026, 5:45:22 PM	Ng Tzi Dong Jeremy	-	Forum: Participation Centre	Forum	Course module viewed	The user with id '109125' viewed the 'forum' activity with course module id '3997247'.	web	[redacted]

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Advanced methods

- Recap: **Eye-tracking research** results show that users' visual attention focuses on specific regions of interest, and their viewing behaviours are not *linear*
- Researchers have also started exploring **neuroscientific methods**, i.e., recording and modelling **brain signals**

https://measuringu.com/eye-tracking/

https://www.nhs.uk/conditions/electroencephalogram/

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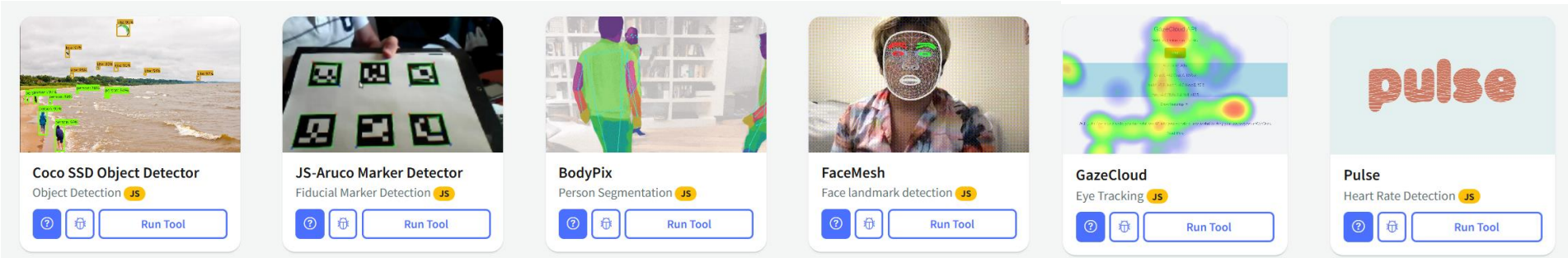
In-class activity: Multimodal user evaluation

- With advancement of analytics and AI, more toolkits have been developed to collect **multimodal data** for user-centered evaluation of websites.
- EZ-MMLA toolkit: developed by Learning, Innovation and Technology lab at the Harvard Graduate School of Education

https://mmla.gse.harvard.edu/tools/

- Task: Explore the toolkit by trying a few tools (e.g., *GazeCloud, FaceMesh*). Think about how the tools can help you evaluate websites:



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Outline

- Recap
- Usability
- User-Centered Evaluation
- **Preparation for assignments: Group Project**

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Take-home task: Please form a group (Due: Feb 17)

- Form a group of 3 – 4 students.
- The entire group should "enter" a group on Moodle:

Choice	Group	Show descriptions	Members / Capacity	Group members	Show Group Members
○	Group A		0 / 4		
○	Group B		0 / 4		
○	Group C		0 / 4		
○	Group D		0 / 4		
○	Group E		0 / 4		
○	Group F		0 / 4		
○	Group G		0 / 4		
○	Group H		0 / 4		

https://moodle.hku.hk/mod/choicegroup/view.php?id=4043545

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Outline

- Recap
- Usability
- User-Centered Evaluation
- **Preparation for assignments: Individual Project**

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3. Website creation and presentation (35%) (Deadlines: TBC)

As an individual project, each student will create a new website using tools/platforms introduced in class and/or those they are familiar with. This individual project is comprised of a **publicly accessible website** (15%), a **documentation** (10%), and a 3-**minute pitch presentation** (10%). Detailed instructions and assessment criteria will be announced in Week 8 and will be uploaded to Moodle soon. Please stay tuned.

- As emphasized, *before* (and *during*) website creation, students need to conduct **user studies**, e.g., user needs assessment, where we ought to be **ethical**.
- (More details later) User needs assessment: To elicit needs from 5 – 10 target users through *open-ended* surveys and/or interviews.
- ALL surveys, interviews and/or *any* human data collection procedures can only be conducted ***after*** getting users' **informed consent.**

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Individual project: *Informed Consent Form*

- Please use this template (on Moodle): BSIM3021 Informed Consent Form
  https://moodle.hku.hk/mod/resource/view.php?id=4034955

- If you are conducting a questionnaire survey, place the form at the beginning of the questionnaire
- If you are conducting an interview, please ask your users to sign it first, *before* starting to ask the interview questions.

- For the individual project, you are **required** to submit proof of users' informed consent, e.g., Google Form responses, scanned filled forms, etc. (More details in later week)

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Summary

- Usability
  - What is Usability?
  - Why is Usability important?
  - How to improve Usability?
- User-Centered Evaluation
  - 5-second Usability Test
  - Task analysis; Log analysis; Advanced methods and multimodal user evaluation
- Preparation for assignments:
  - Group Project: guidelines
  - Individual Project: informed ethical consent form

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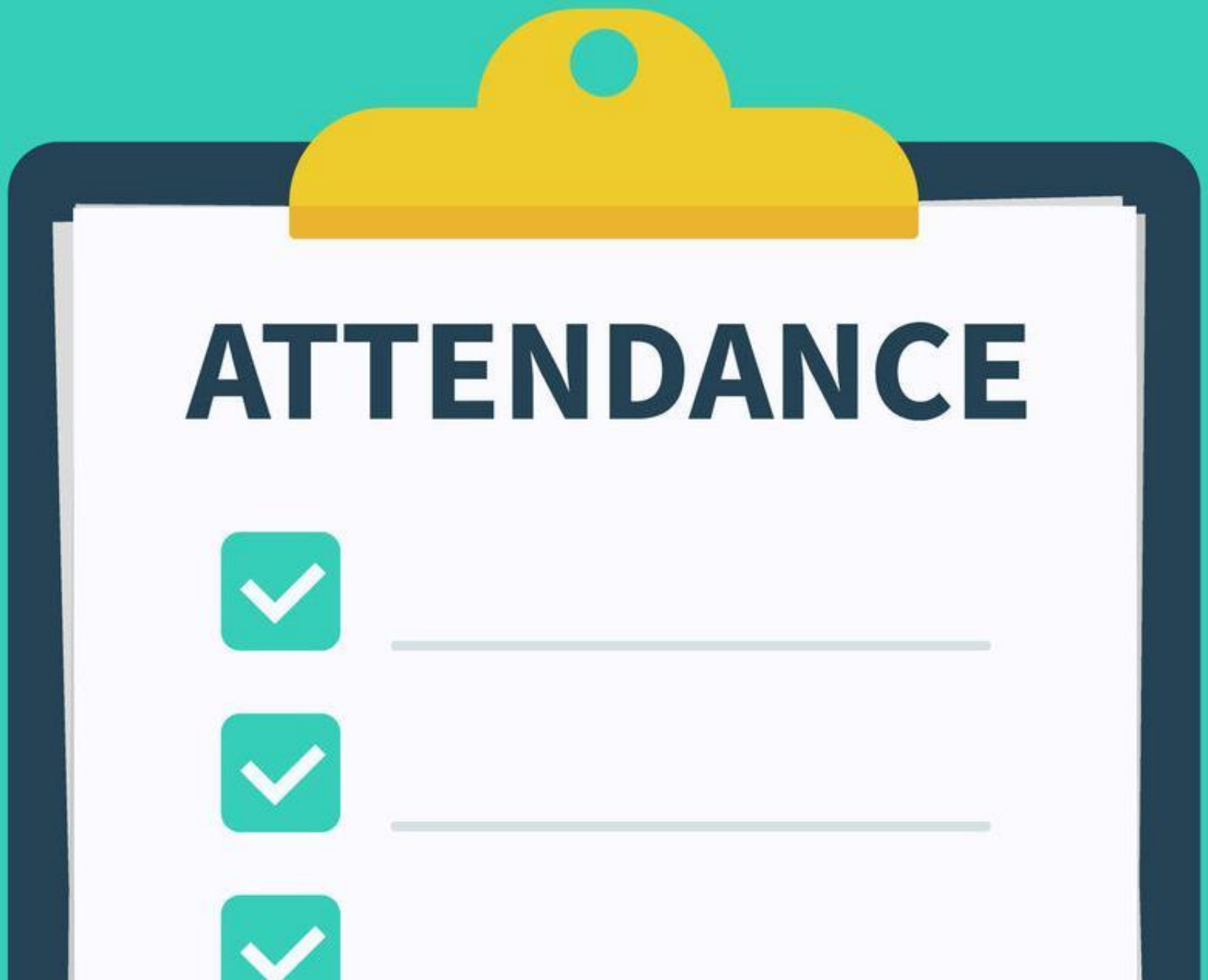
Any
QUESTIONS?


ATTENDANCE

Preview of next week: Feb 24 (Wed)

- Web Accessibility

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